



Oral and Gut Microbiome Alterations in Oral Chronic GVHD Disease: Results from Close Assessment and Testing for Chronic GVHD (CATCH Study)

Armin Rashidi^{1,2}, Joseph Pidala³, Betty K. Hamilton⁴, Steven Z. Pavletic⁵, Katie Kim⁶, Alex Zevin⁶, Jacqueline W. Mays⁷, and Stephanie J. Lee^{1,2}

ABSTRACT

Purpose: Whether and how the oral microbiome and its changes in allogeneic hematopoietic cell transplantation (alloHCT) recipients may contribute to oral chronic GVHD (cGVHD) pathogenesis is unknown. In addition, although the oral and colonic microbiota are distinct in healthy adults, whether oral microbes may ectopically colonize the gut in alloHCT patients is unknown.

Experimental Design: To address these knowledge gaps, longitudinal oral and fecal samples were collected prospectively in the multicenter Close Assessment and Testing for Chronic GVHD study (NCT04188912). Through shotgun metagenomic sequencing of the samples collected at baseline, oral cGVHD onset, first post-cGVHD onset visit, and 1-year post-HCT time points in patients with oral cGVHD (cases; $N = 29$) or without any cGVHD (controls; $N = 51$), we examined whether (i) oral and/or gut microbiomes and

their longitudinal trajectories differ between cases and controls and (ii) oral and gut microbiomes overlap in alloHCT recipients, especially those developing cGVHD.

Results: A total of 195 samples were analyzed. The onset of oral cGVHD was characterized by an expansion of *Streptococcus salivarius* and *Veillonella parvula* in the oral microbiome. High levels of oral/gut microbiota overlap were observed, particularly in patients with oral cGVHD, suggesting ectopic colonization of the gut by oral bacteria.

Conclusions: The unusual coalescence of two distant niches in these patients may result in short- or long-term consequences for the host, a novel avenue for future research. In addition, this study suggests a contribution of the oral microbiome to oral cGVHD pathogenesis.

Introduction

Chronic GVHD (cGVHD) affects up to one-third of allogeneic hematopoietic cell transplant (alloHCT) recipients, with a median time to onset of ~6 months after HCT (1). cGVHD worsens quality of life and is associated with prolonged immunosuppression and higher nonrelapse mortality (2–4). A better understanding of the early biology of the events leading to cGVHD may improve both the prevention and treatment of cGVHD.

Through frequent assessment of alloHCT recipients, starting before HCT and then every 2 months until 1 year post-HCT, we attempted, in the Close Assessment and Testing for Chronic GVHD

study (CATCH study; www.clinicaltrials.gov registration number: NCT04188912), to “catch” the onset of cGVHD clinical manifestations to better understand its early biology. In this Chronic GVHD Consortium multicenter prospective cohort study (5), we collected longitudinal samples from multiple body sites (blood, conjunctival washing, saliva, buccal mucosa, tongue dorsum, oral biopsy, skin, and stool) to evaluate whether profile changes in blood or other bodily fluids and tissues predict cGVHD onset, severity, and organ involvement.

The oral cavity is in direct interface with the external environment, is equipped with a local biofluid, saliva, and is colonized by a distinct local microbiome. The extent to which the oral microbiome changes after alloHCT is not well characterized. Although extensive studies have focused lower in the alimentary tract on the gut microbiome and GVHD, the relationship between the oral and gut microbiomes has not been reported in the post-alloHCT setting. Therefore, the current study focused on elucidating these relationships through prespecified objectives of the present analysis: (i) Are oral and/or gut microbiomes and their longitudinal trajectory different between patients who develop oral cGVHD by 1-year post-HCT and those who do not? (ii) Do oral and gut microbiomes overlap in alloHCT recipients?

Materials and Methods

The CATCH study protocol was approved by a central institutional review board, the Fred Hutchinson Cancer Center (FHCC) Institutional Review Board. The study was conducted in accordance with local regulations and the Declaration of Helsinki. All participants provided written informed consent. The main inclusion criteria were age ≥ 18 years and alloHCT with a risk of cGVHD of $\geq 25\%$. The main exclusion criteria were chronic GVHD risk of

¹Clinical Research Division, Fred Hutchinson Cancer Center, Seattle, Washington. ²Division of Medical Oncology, Department of Medicine, University of Washington, Seattle, Washington. ³Blood and Marrow Transplantation and Cellular Immunotherapy, H. Lee Moffitt Cancer Center and Research Institute, Tampa, Florida. ⁴Blood and Marrow Transplant Program, Department of Hematology and Medical Oncology, Taussig Cancer Institute, Cleveland Clinic, Cleveland, Ohio. ⁵Immune Deficiency Cellular Therapy Program, Center for Cancer Research, National Cancer Institute, National Institutes of Health, Bethesda, Maryland. ⁶Genomics and Bioinformatics Shared Resource, Fred Hutchinson Cancer Center, Seattle, Washington. ⁷Oral Immunobiology Unit, National Institute of Dental and Craniofacial Research, National Institutes of Health, Bethesda, Maryland.

J.W. Mays and S.J. Lee contributed equally to this article.

Corresponding Author: Armin Rashidi, Fred Hutchinson Cancer Center, 1100 Fairview Avenue N, Mailstop M2-B230, Seattle, WA 98109. E-mail: arashidi@fredhutch.org

Clin Cancer Res 2024;30:4240–50

doi: 10.1158/1078-0432.CCR-24-0875

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Translational Relevance

We found evidence for a possible role of oral microbiota in the initial phases of oral chronic GVHD pathogenesis, potentially introducing novel targets for oral chronic GVHD prevention in patients with oral microbiota disruption after transplantation. In addition, ectopic colonization of oral bacteria in the gut as demonstrated in this study may be involved in immunologic and nonimmunologic complications of transplantation such as intestinal GVHD and systemic infections. Most microbiome studies in the field focused on a single niche (e.g., oral or fecal). The results presented herein suggest the link between the two niches may also be of clinical relevance and warrants further investigation.

≤25% (e.g., umbilical cord blood, bone marrow graft with post-transplant cyclophosphamide, or use of antithymocyte globulin, alemtuzumab, or *ex vivo* T-cell depletion), active disease at time of HCT, HCT-comorbidity index >4, and prior allogeneic HCT. Then, formal study assessments occurred at enrollment (before conditioning initiation), day 60 to 104 post-HCT, and then every other month until 1 year post-HCT, as well as at the time of cGVHD diagnosis. Oral cGVHD was diagnosed according to the NIH Consensus Conference criteria (6, 7).

We included microbiome samples collected at the following time points: T0 (baseline; before conditioning initiation), T1 (onset of oral cGVHD), T2 (the first sample after the onset of oral cGVHD), and T3 (1-year post-HCT). If the first sample after the onset of oral cGVHD was collected at 1 year post-HCT, it was considered a T3 sample. All oral samples were collected on Puritan sterile foam-tipped swabs by brushing the bilateral buccal mucosa or dorsal tongue for 20 seconds and placed in individual RNAlater-filled sterile tubes for transport. Next, fecal samples were collected by patients at home using the same type of swabs, placed in sterile preservative-free tubes, and brought to the clinic. All samples were transferred to -80°C as soon as possible. Cases for the present analysis were defined as patients with oral cGVHD onset by 1-year post-HCT. Controls were defined as patients without cGVHD until 1 year post-HCT. Patients with cGVHD but without oral involvement were not included because of our focus on oral cGVHD. No other inclusion or exclusion criteria were applied for the present analysis.

Then, we evaluated our central repository at FHCC on January 4, 2024, and all eligible samples were retrieved for DNA extraction and sequencing. At the time of data cutoff for this analysis, 177 patients were enrolled in the CATCH study. Eighty of these patients provided samples that met our inclusion criteria for the present analysis. Four positive controls (ZymoBIOMICS Microbial Community DNA Standard) were included in DNA extraction, library preparation, and sequencing. DNA was extracted using the DNeasy PowerSoil DNA isolation kit (QIAGEN). Sequencing libraries were prepared using the Nextera XT DNA Library Preparation Kit (Illumina) scaled down to 1/4 of the standard reaction volume. Then, sequencing libraries were pooled and sequenced on an Illumina NovaSeq 6000 using an S2-300 flow cell and a PE150 configuration. The total sequencing yield was 3.7B read pairs with a median depth of 11.8M read pairs per library. MetaPhlAn4 was used for species-level taxonomic assignment. Default parameters

were used. Next, species with a relative abundance of >0.01 in at least 10% of the samples were selected for further analysis. Taxonomic tables of oral samples (buccal mucosa and tongue dorsum) for each patient were averaged to yield the “oral” microbiota for that patient. All included patients provided both oral samples. α -Diversity (i.e., within-sample diversity) was quantified by the Shannon index (8), which considers the richness and evenness of taxa. β -Diversity (i.e., between-sample diversity) was quantified by Aitchison distance (9), which uses centered log-ratio (CLR)-transformed taxa abundances and accounts for compositionality of microbiome data. Ordination was visualized by principal coordinate analysis using the first two axes, and differences in microbiome composition were statistically examined using an Adonis test with 999 permutations. Differential abundance analysis was performed using Maaslin2 (10). Default parameters were used except the following: standardize = FALSE; normalization = “CLR”; transform = “NONE”; and max_significance = 0.10. The significance threshold was defined for corrected *P* values (FDR method). Within-subject oral-gut microbiome overlap was quantified by the presence/absence-based Jaccard index, which quantifies the intersection between two given samples divided by their union (11). A species-level microbiota heatmap visualizing the results of unsupervised hierarchical clustering was generated using CLR abundances and a ward.D function.

As we included all available samples, no power or sample size calculation was done *a priori*. The sample size (both for biospecimens and patient number) was considered adequate for a pilot study aimed at hypothesis generation. *P* values <0.05 and, in the case of multiple comparisons, FDR-corrected *P* values (P_{adj}) <0.10 were considered statistically significant unless otherwise specified. R 4.2.0 was used for all analyses.

Data availability

The sequencing data reported in this article are available from the NCBI Sequence Read Archive under BioProject ID PRJNA1088546. Any additional information required to reanalyze the data reported in this article is available from the corresponding author (Armin Rashidi, arashidi@fredhutch.org) upon request.

Results

We analyzed 114 samples from 29 cases and 81 samples from 51 controls. Patient characteristics are shown in **Table 1**. Supplementary Table S1 shows the representativeness of study participants. Among our 29 cases, cGVHD severity was mild in 25 patients and moderate in 4. A breakdown of the samples in each group and at each time point is provided in **Table 2** and **Fig. 1**. Because of our preplanned inclusion/exclusion criteria, we did not have any cGVHD cases without oral involvement. Of the 29 cases, oral cGVHD was diagnosed before 1 year post-HCT in 18 patients and first diagnosed at 1 year post-HCT in the remaining 11. Eleven of the 18 patients with oral cGVHD diagnosis before 1 year post-HCT achieved a complete response to oral cGVHD treatment by 1 year post-HCT. First-line treatment of oral cGVHD, including those diagnosed before or at 1 year post-HCT, were observation ($n = 5$), topical steroids ($n = 5$), topical sirolimus ($n = 4$), topical steroids/topical sirolimus ($n = 4$), systemic tacrolimus ($n = 3$), systemic steroids ($n = 3$), systemic steroids/systemic calcineurin inhibitor ($n = 2$), topical steroids/topical calcineurin inhibitor ($n = 1$), systemic/topical steroids ($n = 1$), and systemic steroids/topical sirolimus ($n = 1$). The median onset of oral cGVHD was 188 days (range,

Table 1. Patient characteristics.

Characteristic	Case	Control	Total
<i>N</i>	29	51	80
Age at transplantation, years/median (range)	65 (46–77)	60 (21–73)	62 (21–77)
Sex, <i>n</i> (%)			
Male	17 (59)	31 (61)	48 (60)
Female	12 (41)	20 (39)	32 (40)
Underlying disease, <i>n</i> (%)			
Acute leukemia	16 (55)	22 (43)	38 (47)
Myelodysplastic syndromes	8 (28)	15 (29)	23 (29)
Myeloproliferative diseases	4 (14)	10 (20)	14 (18)
Other	1 (3)	4 (8)	5 (6)
Conditioning intensity, <i>n</i> (%)			
Myeloablative	12 (41)	26 (51)	38 (47)
Reduced intensity/nonmyeloablative	17 (59)	25 (49)	42 (53)
GVHD prophylaxis, <i>n</i> (%)			
CNI/MTX based	13 (45)	18 (35)	31 (39)
PTCy/CNI based	2 (7)	15 (29)	17 (21)
CsA + sirolimus + MMF	10 (34)	9 (18)	19 (24)
CNI + MMF	4 (14)	4 (8)	8 (10)
CsA + sirolimus	0	3 (6)	3 (4)
Sirolimus + MMF	0	2 (4)	2 (3)
Center, <i>n</i>			
Fred Hutchinson Cancer Center	22 (76)	30 (59)	52 (65)
Moffitt Cancer Center	3 (10)	13 (25)	16 (20)
Cleveland Clinic Cancer Center	4 (14)	8 (16)	12 (15)

Abbreviations: CNI, calcineurin inhibitor; CsA, cyclosporin A; MMF, mycophenolate mofetil; MTX, methotrexate; PTCy, posttransplant cyclophosphamide.

110–340) after HCT, by design reflecting the timing of T1 samples. T2 samples were collected at a median of 260 (range, 170–310) days after HCT. Antecedent acute GVHD with intestinal involvement (all stage 2) was present in three cases.

Species-level taxonomic assignment mirrored the information provided by the manufacturer and was consistent among the four positive controls (Supplementary Fig. S1). The median, mean, and IQR for the number of species in patient samples were 46, 59, and 2 to 192, respectively. The distribution of the 11 most abundant genera in each sample is shown in Fig. 2A and B.

Table 2. Temporal distribution of samples in the two groups.

	Case	Control
Total, <i>N</i>		
Patients	29	51
Samples	114	81
Oral	59	61
Stool	55	20
T0 (baseline), <i>N</i>	42	67
Oral	23	49
Stool	19	18
T1 (oral cGVHD onset), <i>N</i>	34	NA
Oral	16	
Stool	18	
T2 (first after oral cGVHD onset), <i>N</i>	17	NA
Oral	9	
Stool	8	
T3 (1-year posttransplant), <i>N</i>	21	14
Oral	11	12
Stool	10	2

Microbiome differences between cases and controls, and among time points

Oral and gut microbiota compositions were compared between groups and time points in Fig. 2C and D. Because the baseline microbiome (both oral and gut) was not significantly different between cases and controls (Supplementary Fig. S2A and S2B), in further analysis, T0 samples from cases and controls were grouped together. Each cluster in principal coordinates analysis (Fig. 2C and D) shows samples from one of the groups at one of the time points, with oral samples being in Fig. 2C and fecal samples in Fig. 2D. These clusters for the gut microbiome did not show significant separation (i.e., no significant differences in gut microbiome composition; overall $P = 0.083$; Fig. 2D), and the comparison for oral microbiome was highly significant among clusters (overall $P = 0.002$; Fig. 2C). After adjustment of pairwise P values because of multiple clinically relevant comparisons and using a predefined P_{adj} threshold of 0.10 for statistical significance, T0 and T2 oral microbiomes were significantly different ($P = 0.011$; $P_{\text{adj}} = 0.066$). The T1 ($P = 0.048$; Supplementary Fig. S2C) but not T2 ($P = 1.00$; Supplementary Fig. S2D) oral microbiome was significantly different from the T3 oral microbiome among controls. A trend was observed in the comparison between T0 and T3 oral microbiomes for both cases ($P = 0.08$; $P_{\text{adj}} = 0.12$) and controls ($P = 0.055$; $P_{\text{adj}} = 0.12$). In addition, a trend was found for different compositions between T0 and T2 oral microbiomes ($P = 0.07$; $P_{\text{adj}} = 0.12$). The T3 oral microbiome was not different between cases and controls ($P = 0.36$; $P_{\text{adj}} = 0.43$). Similarly, T2 and T3 oral microbiomes among cases were not different ($P = 0.70$; $P_{\text{adj}} = 0.70$). In an exploratory analysis of the oral microbiome from cases diagnosed with cGVHD within the first year post-HCT, we examined 11 patients who provided a T3 oral sample. Among these patients, four experienced resolution of cGVHD, while seven had ongoing disease. Our analysis revealed no association between the response to cGVHD treatment and the microbiome composition at the 1-year mark (Adonis $P = 0.36$).

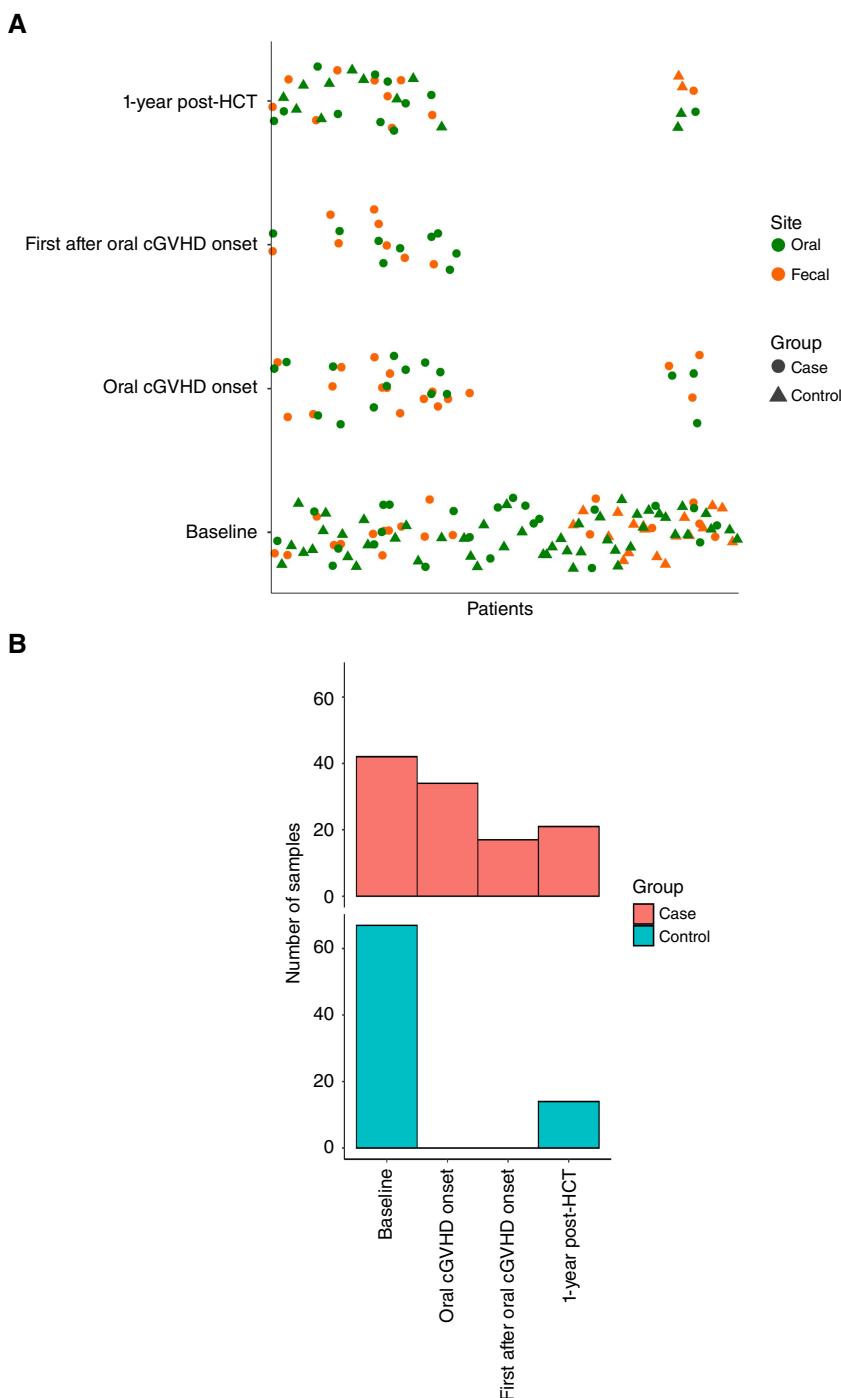


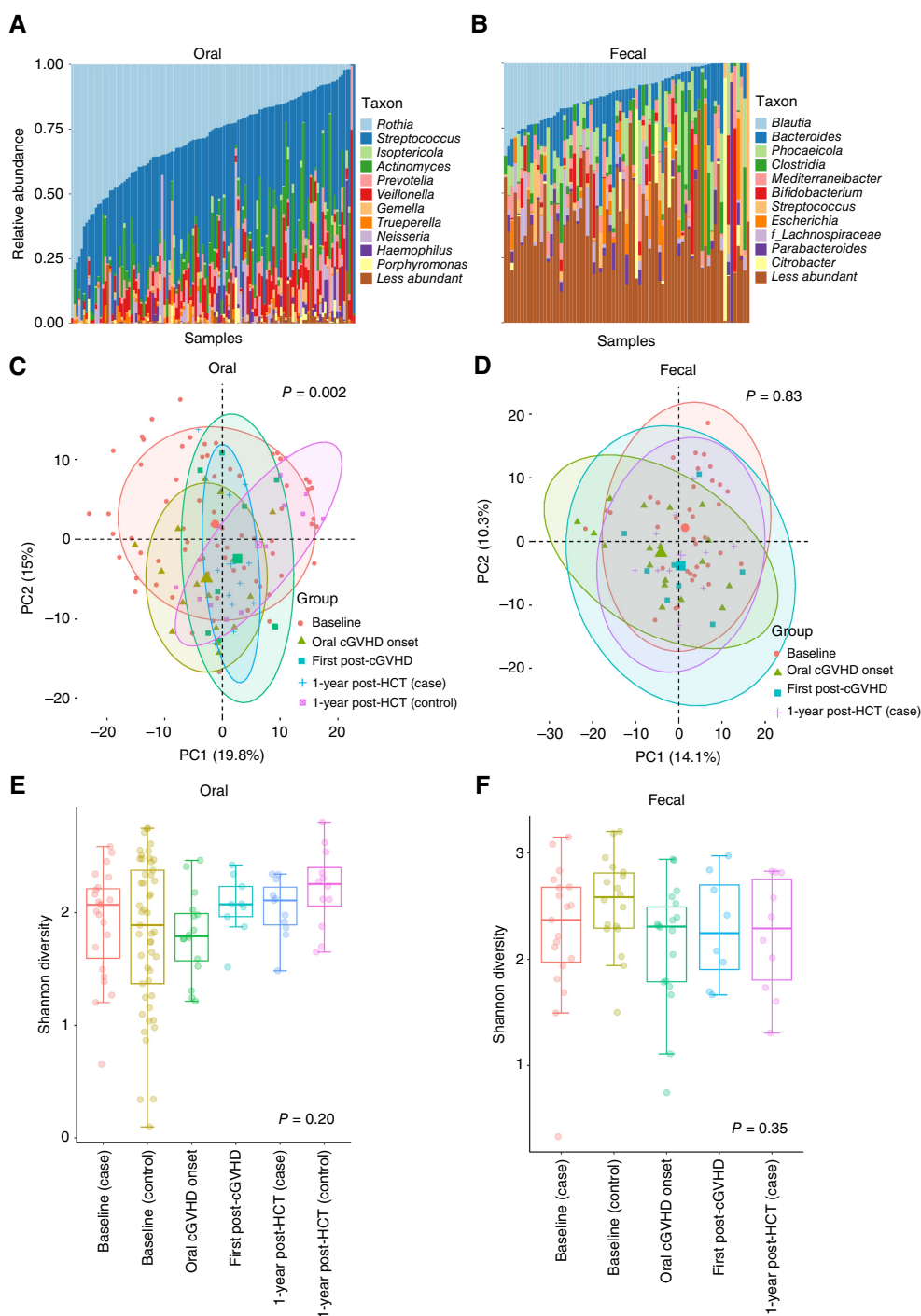
Figure 1.

Sample distribution between groups and among time points. **A**, Each symbol represents a sample. The x-axis shows patients, with each column representing samples from one patient. A small jitter is added to avoid superimposition. **B**, Sample histogram for each group.

A similar pattern, although statistically nonsignificant after adjustment of *P* values for multiple comparisons, was apparent in α -diversity analysis. An initial decline in oral microbiome diversity from baseline to the onset of oral cGVHD (T1) was followed by recovery of diversity until 1 year post-HCT (**Fig. 2E**), consistent with the repeatedly observed resilience of the oral microbiome and its

resistance to durable change (12–14). Additionally, no clear pattern was apparent in gut microbiota diversity. Together, these findings indicated a significant transient change in the oral microbiome at the onset of oral cGVHD, which was largely resolved by 1 year post-HCT.

To further characterize major changes in the oral microbiome at T1 versus T0, we performed differential abundance analysis

**Figure 2.**

Microbiota composition and diversity analysis. **A** and **B**, Distribution of the 11 most abundant taxa in each sample. Less abundant taxa were combined in one category. Each column is a sample, and its corresponding stacked bars indicate relative abundances of different taxa in that sample. **C** and **D**, β -Diversity and ordination visualized by principal coordinate analysis. Aitchison distance (using CLR-transformed species abundances) was used to quantify the overall compositional difference between samples. The first two principal coordinates (PC1 and PC2) are shown. Numbers in parentheses indicate percent variation explained by the corresponding axis. P values are from an Adonis test with 999 permutations. Each symbol represents a sample. The closer the two samples, the more similar their microbiome composition. The centroid of each cluster is shown by a larger symbol. Ninety-five percent ellipses are shown. Too few T3 samples were found among controls to make a cluster in **D**. Because the baseline microbiome (both oral and gut) was not significantly different between cases and controls (Supplementary Fig. S2), T0 samples from cases and controls were grouped together. **E** and **F**, α -Diversity analysis using Shannon index. Each circle represents a sample. Each box shows median (horizontal middle line) and IQR. Whisker lines indicate non-outlier maximum and minimum values. A small jitter is included for better visualization.

comparing these two time points. In this analysis, the relative abundance of *Veillonella parvula* and *Streptococcus salivarius* was significantly higher at T1 than T0 ($P_{\text{adj}} = 0.06$ for both species; Fig. 3A and B), indicating their expansion at the onset of oral cGVHD. Supplementary Table S2 shows the results for other species, none of which were statistically significant. To further evaluate whether change in the oral microbiome at the onset of oral cGVHD might durably destabilize the microbiome, we quantified the change in microbiome composition between T0 and T3 in cases versus

controls, using species presence/absence data and Jaccard distance. In this analysis, the overall extent of microbiome change between T0 and T3 seemed to be greater in cases than controls, a trend that did not reach statistical significance ($P = 0.18$; Fig. 3C).

Oral and gut microbiota overlap

To address the second preplanned objective of this analysis—oral-gut microbiome overlap—we first visualized a microbiota heatmap of oral versus fecal samples (Fig. 4). This unsupervised

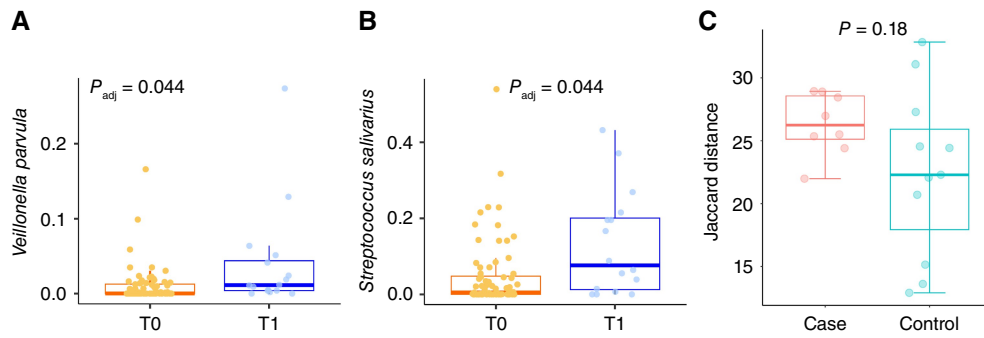


Figure 3.

Significant results from differential abundance analysis and longitudinal oral microbiome change. **A** and **B**, Two species were differentially abundant between T0 and T1 time points in the oral microbiome of cases. Their relative abundances at each time point are shown. Each circle represents a sample. Each box shows median (horizontal middle line) and IQR. Whisker lines indicate nonoutlier maximum and minimum values. A small jitter is included for better visualization. P_{adj} values are from Maaslin2 analysis which applied FDR correction to raw P values because of multiple testing. **C**, Species-level presence/absence-based dissimilarity (i.e., Jaccard distance) of baseline and 1-year posttransplant oral microbiome. Each circle represents a sample. Each box shows median (horizontal middle line) and IQR. Whisker lines indicate nonoutlier maximum and minimum values. A small jitter is included for better visualization. P value is from a Wilcoxon test.

hierarchical clustering demonstrated a clear distinction between the two niches. However, it also showed substantial oral/fecal overlap based on the presence of multiple taxa in oral and fecal samples, motivating more in-depth analysis to examine within-subject overlap. To this end, we applied Jaccard distance and quantified the fractional overlap between gut and oral microbiome within the same individual and at the same time point. This index was calculated for all such sample pairs. Next, in sharp contrast to the case in healthy adults (15–18), we found a significant overlap between oral and gut microbiota in almost all patients, with a median Jaccard distance of 0.89 (as opposed to ~ 1 in healthy adults) and a range of 0.41 to 1.00. This distance was significantly greater in controls than cases ($P = 0.04$; **Fig. 5A**). Importantly, even at baseline the median distance was < 0.9 and not higher than post-HCT time points (**Fig. 5B**), indicating the effect of prior events and exposures.

Finally, we characterized the species with large contributions to within-individual overlap between oral and gut microbiota. Overall, the same species were found in both cases and controls, with the two dominant species being *Streptococcus parasanguinis* and *S. salivarius*, each overlapping between oral and gut microbiome of 10% to 15% of both cases and controls (**Fig. 5C and D**). Similar to our prior work (16, 19), we reasoned that for an overlapping species to have originated from the mouth, it should demonstrate a higher relative abundance in the mouth (its natural habitat) than in the colon. Visual examination of the plots in **Fig. 5E and F** demonstrates this pattern in all overlapping species except *Bifidobacterium longum*. This species overlapped in the gut and oral microbiome of two patients, with a higher relative abundance in the stool than mouth, arguing against an oral origin. *Bifidobacterium longum* is one of the earliest colonizers of the colon in infants (20), the most common *Bifidobacterium* in the gut of children and adults (21), and is a probiotic (22).

Discussion

This work represents the first longitudinal analysis of the oral and gut microbiome in the context of oral cGVHD. Choosing the control group as alloHCT patients without cGVHD versus healthy adults allowed us to distinguish cGVHD-related changes in the oral

microbiome from the more general HCT-related changes. In addition, shotgun metagenomic sequencing permitted species-level resolution, an important step toward potential precision diagnostic and therapeutic trials. The relationship between the microbiota and cGVHD has been studied in previous work. Changes in the gut microbiota can alter the risk of cGVHD via mechanisms involving short-chain fatty acids (23, 24) and regulatory T cell to Th1 ratio (25). Additionally, oral dysbiosis at baseline has been associated with increased relapse risk after alloHCT (26). To the best of our knowledge, only one prior study evaluated the association between the oral microbiome and oral cGVHD (12). In this study, microbiome diversity and composition were not associated with oral cGVHD. The study used 16S rRNA short-amplicon sequencing, and 15 patients developed oral cGVHD. Understanding microbiota-outcome associations is important because they can potentially lead to microbiota-targeted interventions to prevent adverse events after alloHCT (27).

We demonstrated a significant change in the oral microbiome at the onset of oral cGVHD, characterized by an expansion of *V. parvula* and *S. salivarius*. There are two explanations for this finding. First, the change may have already started before oral cGVHD onset. This scenario could indicate the potential contribution of oral dysbiosis to cGVHD pathogenesis. The role of oral dysbiosis in the pathogenesis of patients with primary Sjögren syndrome and dry mouth was suggested in a previous comprehensive microbiome analysis (28). The baseline oral microbiome was not different between cases and controls, effectively ruling out a contribution by the baseline microbiome to future oral cGVHD risk. However, post-transplant oral dysbiosis may have initiated or worsened the pathogenetic processes culminating in oral cGVHD. Our findings would suggest this microbial stimulus was transient as it was no longer detected at 1 year post-HCT. Alternatively, oral microbiota change (apparent at the time of sampling) may have been a consequence of cGVHD. By providing nutrients for oral microbes and protecting them against colonization by nonoral microbes, saliva exhibits an important role in microbial ecology of the oral cavity. Dry mouth, a hallmark finding in oral cGVHD, may have thus disrupted salutary effects of saliva and led to microbiota alterations. This has been well established in patients with primary Sjögren syndrome, in which



Figure 4.

Microbiota heatmap. Species-level microbiota heatmap visualizing the results of unsupervised hierarchical clustering using CLR-transformed taxa abundances and a ward.D function. CLR values show the ratio of each taxon to the (geometric) mean of all other taxa. Each column is a sample, and each row is a species. The blue-red gradient shows species abundances scaled rowwise. Sites are added along the top border.

reduced salivary secretion explains more than 90% of microbiota variation among patients (29). The observed changes were largely resolved by 1 year post-HCT, suggesting a cGVHD effect on the oral microbiome, which was reversed with successful cGVHD treatment. In a comprehensive study of the microbiota in patients with primary Sjögren syndrome, response to hydroxychloroquine treatment was associated with the alleviation of oral microbiome alterations (28). The relatively modest number of cases in our study did not permit a meaningful microbiome analysis with regard to cGVHD response to treatment. In limited exploratory analysis, however, we found no association between treatment response and microbiome composition. Therefore, we slightly favor the first model, in which a transient change in the oral microbiome of unclear etiology triggers or fuels the immunologic pathways leading to oral cGVHD.

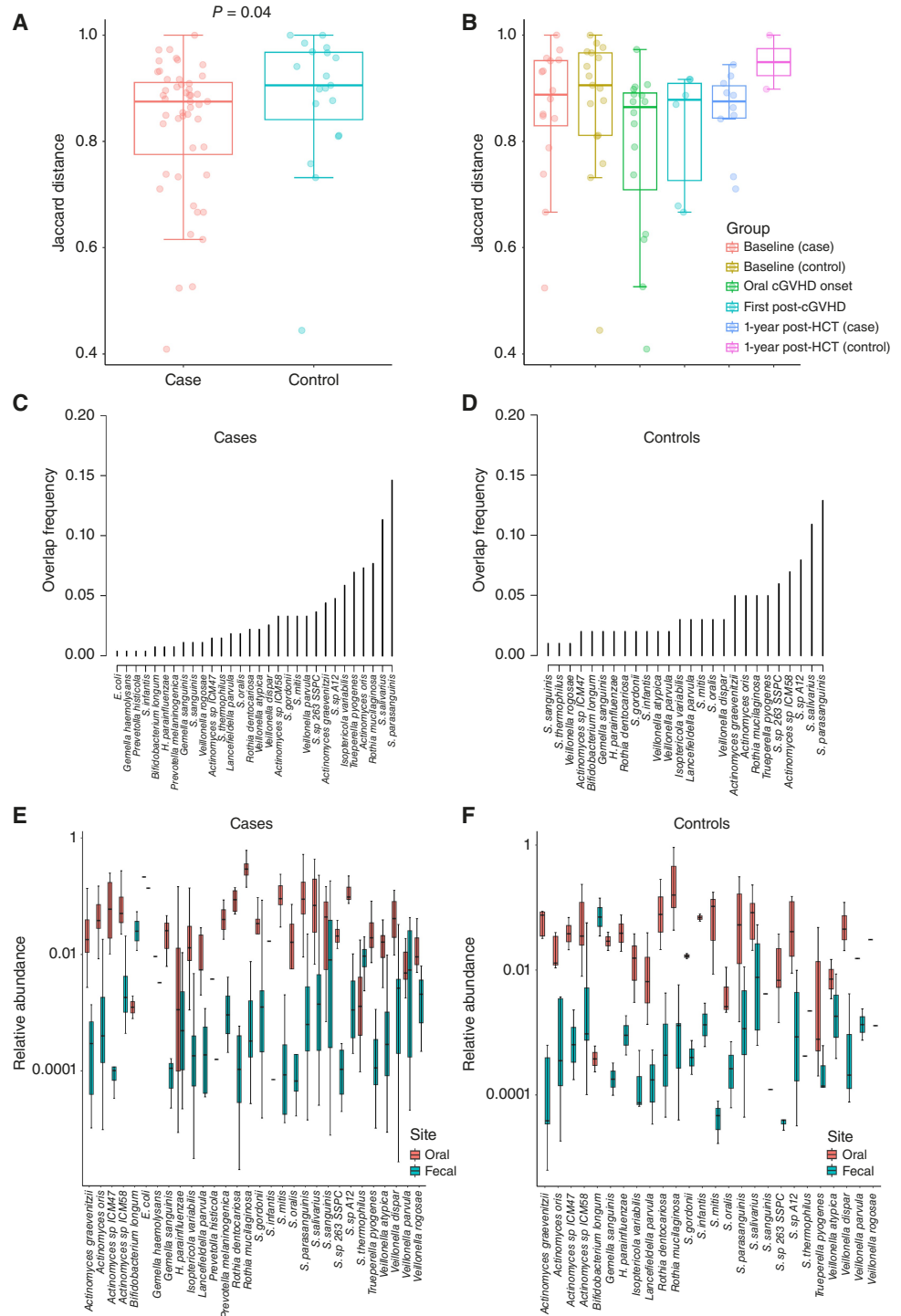
V. parvula is an anaerobic Gram-negative member of the normal oral flora and is involved in early pathogenesis of periodontal disease (30). *V. parvula* stimulates the release of TNF- α and IL6 in human peripheral blood mononuclear cells (31), via a toll-like receptor 4-dependent pathway. In contrast, *S. salivarius* inhibits immune activation by periodontopathogens (32, 33). This function

seems to be mediated at least partly via its surface molecule G7 lipoteichoic acid inhibiting the binding of LPS and lipoteichoic acid of periodontopathogens to CD14 and LPS binding protein, which are required for toll-like receptor activation. Therefore, the mechanisms by which cGVHD may influence these two species or vice versa need further research. Next, disruptions to the dental plaque and salivary microbiota have been associated with the risk of acute GVHD (34, 35), raising the possibility of contribution by oral dysbiosis to immunological complications of HCT. The seemingly transient nature of oral microbiota change at the onset of oral cGVHD is consistent with the known resilience of the oral microbiome and its resistance to durable change (13, 14).

Our second question in this work, namely the within-subject overlap between oral and gut microbiomes, was inspired by our previous analyses (15, 16) and those by others (17, 18) showing that despite swallowing 1 to 1.5 L of saliva every day, oral and gut microbiomes in healthy adults exhibit almost no overlap, with *S. salivarius* and *S. parasanguinis* as possible exceptions. The reasons for this distinct niche specificity have not been ascertained but gastric acidity, low oxygen pressure in the colon, and colonization resistance

Figure 5.

Gut/oral microbiota overlap. **A** and **B**, Within-subject gut/oral microbiota compositional overlap in different groups, at different time points, and using species-level presence/absence data and Jaccard distance. **A**, Jaccard distance value of 1 means no overlap, which is the case for most healthy adults. All time points are combined in **A**, whereas **B** shows them separately. Each circle represents a sample pair (i.e., oral and fecal samples from the same patient at the same time point). Each box shows the median (horizontal middle line) and IQR. Whisker lines indicate nonoutlier maximum and minimum values. A small jitter is included for better visualization. P value is from a Wilcoxon test. **C** and **D**, Frequency of gut/oral microbiota compositional overlap on a per-patient basis and using species-level presence/absence data. Each bar shows the proportion of sample pairs in which the corresponding species was present in both samples. **E** and **F**, Oral and fecal relative abundances of overlapping species. Two boxes are shown for each species, one showing its oral relative abundance and the other one showing its fecal relative abundance. Each box shows median (horizontal middle line) and IQR. Whisker lines indicate non-outlier maximum and minimum values.



of the gut microbiota may be involved in opposing the successful arrival and survival of oral bacteria in the distal gut. Here, we asked whether the same scenario is true in alloHCT recipients. Several major alterations occur in niche-protective mechanisms listed previously, including the frequent use of proton pump inhibitors (decreasing gastric acidity), conditioning and GVHD-induced intestinal inflammation and hyperemia (increasing mucosal and, ultimately,

luminal oxygen pressure), and antibiotic pressure (decreasing colonization resistance). Therefore, we hypothesized that a completely different situation, with a significant overlap between gut and oral microbiomes, may be present in alloHCT patients. Our preliminary data from genus-level analysis of a public database suggested species of the genus *Streptococcus* may co-colonize oral and gut microbiomes in these patients (19). However, because our methodology did not

offer species-level resolution, we could not determine whether they were the same species (i.e., consistent with an oral origin and ectopic colonization in the gut via saliva) or different species of the same genus (i.e., consistent with niche specialization of different species unrelated to anatomic connections). Our shotgun approach in the present analysis allowed us to address this question.

We demonstrated a substantial within-subject overlap between oral and gut microbiome in most patients. Importantly, the overlapping species were of oral origin, thus suggesting a salivary route for ectopic colonization in the colon. Interestingly, the top 2 overlapping species (*S. salivarius* and *S. parasanguinis*) were identical between alloHCT patients and healthy adults. This finding suggests the intrinsic ability of these two species to resist the multitude of physicochemical barriers along the gastrointestinal tract and colonization resistance of the gut microbiota when these barriers are weakened. As an example, gastric pH ranges between 1.5 in the fasting state and 3.0 to 5.0 during eating (36). *S. salivarius* is one of the few oral bacteria that, because of its urease-producing activity, can increase the pH in their microenvironment (37), allowing it to resist gastric acid. AlloHCT is associated with impairments in important barrier mechanisms such as gastric acid (e.g., use of proton pump inhibitors), gut colonization resistance (e.g., antibiotic exposures), and niche-specific ecological properties (e.g., loss of gut luminal hypoxia caused by intestinal mucositis and acute GVHD), providing a perfect setting for ectopic colonization by specific oral bacteria. Importantly, this phenomenon was present even at baseline, indicating that prior events such as those during chemotherapy for the underlying malignancies were also contributory. This is not surprising because many of the events disrupting physiologic barriers also occur during chemotherapy for hematologic malignancies, including intensive chemotherapy for acute leukemia. Furthermore, the extent of overlap in our cases was greater than controls. Several explanations can be entertained. Change in the oral microbiome at the onset of oral cGVHD may be associated with expansion or emergence of more resistant strains that are more likely to reach and colonize the gut. Alternatively, concurrent intestinal inflammation or antibiotic exposures may have weakened gut colonization resistance more extensively in cases than controls. Finally, medications to treat cGVHD may have influenced some of the barriers between mouth and colon in cases, facilitating ectopic colonization. The implications of ectopic colonization of oral bacteria in alloHCT patients merit further research. Large overlaps between oral and colonic flora have been reported in patients with acute leukemia undergoing induction chemotherapy (38), potentially explaining the overlap we observed in our patients at baseline. In patients with inflammatory bowel disease, this phenomenon is known to increase disease activity and severity (39–43).

One limitation of this study is its pilot design. The objective was hypothesis generation for future research rather than mechanistic inference. Although we used predefined hypotheses and analytic methods to minimize the risk of false discovery, our findings need validation. In addition, we did not have a sample from the control group corresponding to the time point of oral cGVHD onset among cases or a nonbaseline sample before the onset of oral cGVHD among cases. This limitation prohibits a firm conclusion about the causality of microbiome changes in oral cGVHD pathogenesis. Another limitation is related to how the stool samples were collected. Without a DNA preservative, it is possible that samples kept at room temperature for long enough suffered loss of certain taxa.

The somewhat lower fecal α -diversity and its nonsignificant change over time in our patients in contrast to prior reports (44) may be explained by this limitation. Although patients were generally not on antibacterial antibiotics at the time points of sample collection in this work, some of the earlier T1 samples may have not yet recovered from periengraftment antibacterial antibiotics. We do not expect such effects to be prominent on the oral samples because systemic antimicrobials are known to exhibit only a small effect on the composition of the oral bacterial community (owing to the biofilm properties of oral bacteria giving them substantial resistance/resilience; refs. 13, 45, 46). Similarly, the possible effect of antecedent acute GVHD on our findings could not be determined because of small numbers. In addition, we did not have granular data on oral hygiene and tooth loss, which might have influenced some of our findings. Finally, although the concurrent presence of the same species in the mouth and gut and with a higher abundance in the mouth is highly suggestive of ectopic colonization of oral bacteria in the gut, strain-level analysis would be needed to ascertain this.

In conclusion, in this first longitudinal analysis of the oral and colonic microbiome in alloHCT patients with and without oral cGVHD, we found an expansion of *S. salivarius* and *V. parvula* at the onset of oral cGVHD. How these species may contribute to cGVHD pathogenesis requires further research. A temporal and mechanistic link, if established in future validation studies, could provide rationale for microbiota-focused interventions in patients deemed high-risk for future oral cGVHD. Whether the seemingly transient oral microbiome changes demonstrate implications for longer-term oral complications, such as periodontal disease and tooth loss, is unknown. We demonstrated high levels of within-subject oral/gut microbiota overlap (especially those with oral cGVHD), at a higher frequency and magnitude than previously reported in other clinical settings. The consequences of this remarkable inter-niche coalescence for the host remain unknown. Further proteomic and metabolomic studies of saliva supernatants should decipher the mechanisms for our findings.

Authors' Disclosures

A. Rashidi reports support from Genomics & Bioinformatics Shared Resource, RRID: SCR_022606, of the Fred Hutch/University of Washington/Seattle Children's Cancer Consortium, and Fred Hutchinson Cancer Center during the conduct of the study, as well as consulting fees from Seres Therapeutics and Emmes DSMC outside the submitted work. B.K. Hamilton reports other support from MaaT Pharma, ACI Group, Angiocrine, Incyte, Sanofi, and CSL Behring outside the submitted work. S.Z. Pavletic reports research support from the Center for Cancer Research at the NCI through the NIH Intramural Research Program [includes Clinical Research Development Agreements with Actelion, Celgene, Eli Lilly and Company, Kadmon (now Sanofi), and Pharmacyclics]. J.W. Mays reports being a federal government employee (at the NIH) and reports that this work was done as part of official duties, which impacts the copyright. S.J. Lee reports grants and nonfinancial support from Astra-Zeneca and Pfizer; grants, personal fees, and nonfinancial support from Sanofi; grants from Syndax; nonfinancial support from Janssen; and personal fees from Mallinckrodt, Equillum, Kadmon, Novartis, and Incyte outside the submitted work. No disclosures were reported by the other authors.

Authors' Contributions

A. Rashidi: Conceptualization, data curation, formal analysis, investigation, visualization, methodology, writing—original draft. **J. Pidala:** Writing—review and editing. **B.K. Hamilton:** Writing—review and editing. **S.Z. Pavletic:** Writing—review and editing. **K. Kim:** Project administration, writing—review and editing. **A. Zevin:** Resources, writing—review and editing. **J.W. Mays:** Conceptualization, methodology, writing—review, and editing. **S.J. Lee:** Conceptualization, resources, supervision, funding acquisition, methodology, project administration, writing—review and editing.

Disclaimer

The content is solely the responsibility of the authors and does not represent the official views of the listed organizations. The funders had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The last author had full access to all the data in the study and final responsibility for the decision to submit for publication.

Acknowledgments

We thank the Specimen Processing & Research Cell Bank of FHCC for sample processing and DNA extraction. This research was supported by the Genomics and Bioinformatics Shared Resource, RRID: SCR_022606, of the Fred Hutch/

University of Washington/Seattle Children's Cancer Consortium (P30 CA015704), NCI grant CA118953, and National Institute of Dental and Craniofacial Research grant CA236229. J.W. Mays was supported by the Intramural Research Program of the National Institute of Dental and Craniofacial Research, NIH (ZIA DE000747).

Note

Supplementary data for this article are available at Clinical Cancer Research Online (<http://clincancerres.aacrjournals.org/>).

Received March 17, 2024; revised May 13, 2024; accepted July 15, 2024; published first July 17, 2024.

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